

Uncertainty functionals revisited: Concavity and Jensen's inequality

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Abstract

This article presents a theoretical study of uncertainty functionals on general measurable spaces. These functionals are fundamental in experimental design and global sensitivity analysis, where they are used to quantify variability and information content in probabilistic models. As first articulated in DeGroot's seminal 1962 article, a natural requirement is that uncertainty should decrease on average when additional information is obtained. This requirement is equivalent to the probabilistic form of Jensen's inequality on the space of probability measures. Our main results show that concavity is necessary but not sufficient for Jensen's inequality to hold whenever the underlying measurable space is infinite. We also provide practicable sufficient conditions under which the desired property holds. These results contribute to a clearer mathematical foundation for uncertainty quantification. Several open questions are formulated.

Keywords: Uncertainty functionals, Decreasing on average, Concavity, Jensen's inequality

1 Introduction

The story of uncertainty functionals originates in the late 1950s in the field of Bayesian experimental design, when Lindley (1956, 1957) proposed measuring the information provided by an experiment about a parameter $\theta \in \Theta$ through the difference

$$\Delta = \mathbb{E} \left(I \left(\mathbb{P}^{\theta|X} \right) \right) - I \left(\mathbb{P}^{\theta} \right),$$

where X denotes the result of the experiment, and the “information” $I(\nu)$ carried by a probability measure ν on Θ is defined as the negative of the (differential) entropy with respect to a given reference measure λ :

$$I(\nu) = \int \frac{d\nu}{d\lambda} \log \left(\frac{d\nu}{d\lambda} \right) d\lambda,$$

assuming that $\nu \ll \lambda$. Lindley observes, among other properties, that this difference is always non-negative—indeed, it coincides with the expected Kullback–Leibler divergence from the posterior to the prior distribution, or equivalently to the mutual information between θ and X . Assuming that Θ is finite, and λ is the uniform distribution on Θ , Lindley’s information $I(\nu)$ is the negative of Shannon’s entropy $H(\nu)$, which is a non-negative measure of uncertainty, and Δ can be read as the expected reduction of uncertainty provided by the experiment:

$$\Delta = H(P^\theta) - \mathbb{E} \left(H(P^{\theta|X}) \right). \quad (1)$$

(As is well known, $H(\nu)$ can be negative when H is the differential entropy in general; nevertheless, even in this case, the difference Δ remains non-negative.)

DeGroot (1962) took Lindley’s idea one step further, replacing Shannon’s entropy in (1) by a general uncertainty functional, which we will continue to denote by H in this article. An uncertainty functional in the sense of DeGroot¹ is a (measurable) non-negative mapping defined on the set of probability measures on Θ , which vanishes on Dirac measures and yields a non-negative uncertainty reduction Δ for any possible experiment. Functionals that satisfy the latter property will be called *decreasing on average* as in Bect et al. (2019). For probability measures on finite sets, apart from Shannon’s entropy, the α -entropies of Havrda and Charvát (1967) and the Gini-Simpson index (Simpson, 1949) are examples of uncertainty functionals. For probability measures on $\mathbb{R}$, variance is a prototypical example of such a function, which can be seen as a special case of a more general class of uncertainty functionals called quadratic entropies by Rao (1984). The reader is referred to Bernardo (1979); Rao (1982); DeGroot (1986, 1994); Ginebra (2007); Dawid (2007); Hainy et al. (2014) for related discussions and examples.

Uncertainty functionals also play a role in global sensitivity analysis, where the objective is to assess how uncertainties in a model’s inputs propagate through the model and contribute to the variability of its output (see, e.g., Saltelli et al., 2004; Da Veiga et al., 2021, and references therein). Consider indeed the popular first-order Sobol’ sensitivity indices associated with a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and a random input vector X :

$$S_j = \frac{\text{var}(\mathbb{E}(f(X) | X_j))}{\text{var}(f(X))}, \quad 1 \leq j \leq n,$$

originally introduced by Sobol’ (1993) using the functional ANOVA decomposition of f (Hoeffding, 1948). As observed by Homma and Saltelli (1996), these indices admit

¹DeGroot (1962) actually uses the term “uncertainty function”, which is consistent with the fact that the article focuses on the case where Θ is finite. We will prefer the term “uncertainty functional” in order to emphasize that, in general, the space of probability measures on Θ is infinite-dimensional.

a natural interpretation as the (normalized) expected reduction of uncertainty (1) in the output $Y = f(X)$ that can be achieved by reducing the uncertainty on a single input X_j :

$$S_j = \frac{\text{var}(f(X)) - \mathbb{E}(\text{var}(f(X) \mid X_j))}{\text{var}(f(X))} = \frac{H(\mathbb{P}^Y) - \mathbb{E}(H(\mathbb{P}^{Y|X_j}))}{H(\mathbb{P}^Y)}, \quad (2)$$

where $H(\nu)$ denotes the variance of a probability measure ν . Several authors have taken this reformulation as the starting point for a generalization of Sobol’s variance-based sensitivity analysis, replacing the variance by a broader class of functionals H (Fort et al., 2016; Borgonovo et al., 2021; Fissler and Pesenti, 2023; Straub et al., 2025). The special case where $Y = f(X)$ is discrete, and H is Shannon’s entropy, has been proposed by Krzykacz-Hausmann (2001) and further discussed by Auder and Iooss (2009). As above, to ensure that the right-hand side of (2) defines a sensitivity index in $[0, 1]$, interpretable as a fraction of uncertainty reduction, H must be non-negative and decreasing on average.

Focusing on the case where Θ is a finite set, DeGroot (1962) proves that H is decreasing on average if, and only if, it is concave. The “only if” part is proved by considering a suitable Bernoulli experiment, and that part of DeGroot’s argument remains valid even if Θ is not finite. The “if” part follows from the integral—i.e., probabilistic—form of Jensen’s inequality, using the fact that the set of probability measures on a finite set of cardinal n can be identified with the $(n - 1)$ -dimensional probability simplex, which is a convex subset of $\mathbb{R}^n$. That part does not generalize directly beyond the case of finite sets. Indeed, the set of probability measures becomes infinite-dimensional, and it has been proved by Perlman (1974) that Jensen’s inequality does not hold in general in infinite-dimensional spaces. The main contribution of this article is to prove that concavity is not sufficient in general for H to be decreasing on average.

The paper is organized as follows. Section 2 introduces the key concepts underlying uncertainty functionals, provides their formal definitions, and illustrates these definitions with two well-known families of functionals. In particular, we show that the property of being decreasing on average is equivalent to (the probabilistic form of) Jensen’s inequality on the space of probability measures. Section 3 examines the relationship between concavity and Jensen’s inequality. Our main results, based on explicit counterexamples, show that concavity is necessary but not sufficient for Jensen’s inequality to hold whenever the underlying measurable space is infinite. We also discuss a sufficient condition under which Jensen’s inequality holds. Finally, Section 4 summarizes the main findings of this work and discusses directions for future research.

2 Uncertainty functionals

2.1 Definitions

Let $(\mathbb{Y}, \mathcal{Y})$ denote a measurable space, and let $\mathbb{P} = \mathbb{M}_1^+(\mathbb{Y}, \mathcal{Y})$ denote the space of all probability measures on $(\mathbb{Y}, \mathcal{Y})$. The set $\mathbb{Y}$ corresponds to the output space in the sensitivity analysis setting ($\mathbb{Y} = \mathbb{R}$ in our introductory discussion) and to the

parameter space Θ in the experimental design setting. We are interested in functionals defined on $\mathbb{P}$, possibly taking infinite values, i.e., mappings from $\mathbb{P}$ to $\overline{\mathbb{R}} = [-\infty, +\infty]$. We call such functionals *probability functionals* on $(\mathbb{Y}, \mathcal{Y})$.

Let $\mathcal{F}$ the σ -algebra on $\mathbb{P}$ generated by the evaluation maps $\pi_B : \nu \mapsto \nu(B)$, $B \in \mathcal{Y}$. A random element ν in $(\mathbb{P}, \mathcal{F})$ is called a random measure (see, e.g., [Kallenberg, 2002](#), p.105–106), and we denote by $\bar{\nu}$ its intensity measure, defined by $\bar{\nu}(B) = \mathbb{E}(\nu(B))$. Let $(\Omega, \mathcal{A}, \mathbb{P})$ denote a probability space. A function $\nu : \Omega \rightarrow \mathbb{P}$ is a random element in $(\mathbb{P}, \mathcal{F})$ if, and only if, $(\omega, B) \mapsto \nu(\omega, B) \triangleq (\nu(\omega))(B)$ is a probability kernel from $(\Omega, \mathcal{A})$ to $(\mathbb{Y}, \mathcal{Y})$. With this notation, the intensity measure $\bar{\nu}$ is given by $\bar{\nu}(B) = \int_{\Omega} \mathbb{P}(d\omega) \nu(\omega, B)$. In the rest of the article, a probability functional $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}$ is called measurable if it is $\mathcal{F} / \mathcal{B}(\overline{\mathbb{R}})$ -measurable.

Definition 1. We shall say that a measurable probability functional $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}$ satisfies *Jensen's inequality* (concave version) if, for all random probability measures ν on $(\mathbb{Y}, \mathcal{Y})$ such that $H(\nu)$ is quasi-integrable,

$$H(\bar{\nu}) \geq \mathbb{E}(H(\nu)). \quad (3)$$

(Recall that a random variable Z is called quasi-integrable if $\mathbb{E}(Z^+) \wedge \mathbb{E}(Z^-) < +\infty$.)

Remark 1. The random variable $H(\nu)$ is always quasi-integrable if H is non-negative, which will be assumed for our uncertainty functionals (see [Definition 2](#) below).

Leveraging the correspondence (see, e.g., [Kallenberg, 2002](#), Lemma 1.37) between random measures and probability kernels, we can also characterize probability functionals that satisfy Jensen's inequality in terms of regular conditional distributions. Indeed, given a measurable space $(\mathbb{X}, \mathcal{X})$ and a random element $Z = (X, Y)$ in $(\mathbb{X} \times \mathbb{Y}, \mathcal{X} \otimes \mathcal{Y})$, the regular conditional distribution $\mathbb{P}^{Y|X}$, when it exists, is a random measure on $(\mathbb{Y}, \mathcal{Y})$ defined on the probability space $(\mathbb{X}, \mathcal{X}, \mathbb{P}^X)$. Conversely, given a random measure ν on $(\mathbb{Y}, \mathcal{Y})$, defined on some probability space $(\mathbb{X}, \mathcal{X}, \pi)$, we can set $\Omega = \mathbb{X} \times \mathbb{Y}$, $\mathcal{A} = \mathcal{X} \otimes \mathcal{Y}$, $\mathbb{P}(dx dy) = \pi(dx) \nu(x, dy)$, and then view ν as the regular conditional distribution of $Y : (x, y) \mapsto y$ given $X : (x, y) \mapsto x$. The following result then stems immediately from this correspondence.

Proposition 1. A measurable probability functional H satisfies Jensen's inequality if, and only if, it is decreasing on average: for all measurable spaces $(\mathbb{X}, \mathcal{X})$, for all random elements $Z = (X, Y)$ in $(\mathbb{X} \times \mathbb{Y}, \mathcal{X} \otimes \mathcal{Y})$ such that a regular conditional distribution $\mathbb{P}^{Y|X}$ exists and $H(\mathbb{P}^{Y|X})$ is quasi-integrable,

$$H(\mathbb{P}^Y) \geq \mathbb{E}\left(H\left(\mathbb{P}^{Y|X}\right)\right). \quad (4)$$

Remark 2. A regular conditional distribution always exists if $\mathbb{Y}$ is isomorphic to a Borel subset of $[0, 1]$, which includes in particular Polish spaces (see, e.g., [Kallenberg, 2002](#), Chapters 1 and 5).

Definition 2. We will say that a non-negative probability functional $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}_+$ is an *uncertainty functional* if i) it is measurable, ii) satisfies Jensen's inequality (equivalently, decreases on average), and iii) vanishes on Dirac measures.

This definition calls for a series of comments.

Remark 3. Following DeGroot (1962), we require in Definition 2 that $H(\delta_y) = 0$ for all $y \in \mathbb{Y}$, but we do not require that $H(\nu) = 0$ if and only if ν is a Dirac measure. In other words, an uncertainty functional measures the uncertainty about certain features of an unknown element of $\mathbb{Y}$, not necessarily about the element itself. For instance, for any measurable space $(\mathbb{Y}, \mathcal{Y})$, if $\xi : \mathbb{Y} \rightarrow \mathbb{R}$ is a measurable function, then $H : \nu \mapsto \text{var}_{Y \sim \nu}(\xi(Y))$ is an uncertainty functional (see Section 2.2.2 for further examples in this vein). Similarly, the null functional $H : \nu \mapsto 0$ is an uncertainty functional—albeit a rather uninformative one. In contrast, in the related definition of a measure of diversity, Rao (1982, 1984) requires that $H(\nu) = 0$ if and only if ν is a Dirac measure.

Remark 4. The set $\mathbb{P}_H = \{\nu \in \mathbb{P} \mid H(\nu) < +\infty\}$ is not necessarily convex for an uncertainty functional in the sense of Definition 2. As an example where $\mathbb{P}_H$ is not convex, consider the case where $\mathbb{Y}$ is finite or countable, and H is defined by $H(\nu) = 0$ if ν is a Dirac measure, and $H(\nu) = +\infty$ otherwise. It is easily checked that H is an uncertainty functional, and $\mathbb{P}_H$ is the set of all Dirac measures on $\mathbb{Y}$, which is not convex.

Remark 5. It is important to allow the value $+\infty$ when dealing with uncertainty functionals on general measurable spaces $(\mathbb{Y}, \mathcal{Y})$, if one wants to work with functionals defined on the set $\mathbb{P} = \mathbb{M}_1^+(\mathbb{Y}, \mathcal{Y})$ of all probability measures on $(\mathbb{Y}, \mathcal{Y})$. DeGroot (1962) does not explicitly allow $+\infty$, but this is not a severe restriction since he focuses on the case where $\mathbb{Y}$ is a finite set. For probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, however, the situation is different: for instance, the variance functional is finite only for probability measures with a finite second-order moment. (Alternatively, we could work with real-valued functionals defined on a convex subset of $\mathbb{M}_1^+(\mathbb{Y}, \mathcal{Y})$, which would be slightly more restrictive since $\{H < +\infty\}$ is not necessarily convex, as explained in Remark 4.)

2.2 Some examples

2.2.1 ϕ -entropies

As a first example, consider the case where $\mathbb{Y}$ is a finite or countable set, endowed with the discrete σ -algebra $\mathcal{Y} = \mathcal{P}(\mathbb{Y})$. For a probability measure ν on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$, set $\nu_y = \nu(\{y\})$ for $y \in \mathbb{Y}$. The following proposition provides an interesting class of uncertainty functionals in this setting.

Proposition 2. Let $h : [0, 1] \rightarrow \mathbb{R}_+$ denote a non-negative concave function, such that $h(0) = h(1) = 0$. Then

$$H : \nu \mapsto \sum_{y \in \mathbb{Y}} h(\nu_y) \tag{5}$$

is an uncertainty functional on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$ and, for all random probability measures ν on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$ such that $\mathbb{E}(H(\nu)) < +\infty$, we have the relation

$$H(\bar{\nu}) - \mathbb{E}(H(\nu)) = \sum_{y \in \mathbb{Y}} \left(h(\bar{\nu}_y) - \mathbb{E}(h(\nu_y)) \right), \quad (6)$$

where the sum in the right-hand side is well-defined in $\overline{\mathbb{R}}_+$.

Proof. Obviously, the probability functional (5) is measurable and vanishes on Dirac measures. It remains to show that it satisfies Jensen's inequality. We have:

$$H(\bar{\nu}) = \sum_{y \in \mathbb{Y}} h(\bar{\nu}_y), \quad \mathbb{E}(H(\nu)) = \mathbb{E} \left(\sum_{y \in \mathbb{Y}} h(\nu_y) \right) = \sum_{y \in \mathbb{Y}} \mathbb{E}(h(\nu_y)).$$

For each $y \in \mathbb{N}$, since h is concave, Jensen's inequality on $\mathbb{R}^n$ (see [Theorem 2](#) below) gives $h(\bar{\nu}_y) \geq \mathbb{E}(h(\nu_y))$. Thus, we can write $h(\bar{\nu}_y) = (h(\bar{\nu}_y) - \mathbb{E}(h(\nu_y))) + \mathbb{E}(h(\nu_y))$, where the two terms in the right-hand side are non-negative. Summing over $\mathbb{Y}$ yields:

$$H(\bar{\nu}) = \sum_{y \in \mathbb{Y}} \left(h(\bar{\nu}_y) - \mathbb{E}(h(\nu_y)) \right) + \mathbb{E}(H(\nu)).$$

Jensen's inequality and Eq. (6) follow. $\square$

Uncertainty functionals of this form have been considered for a long time in the literature (usually on finite sets, but the extension to countable sets is straightforward), under various names such as ϕ -entropies ([Burbea and Rao, 1982](#); [Ben-Tal and Teboulle, 1986](#)) or trace-form generalized entropies ([Scarfone, 2013](#); [Tempesta, 2015](#)). Important special cases are obtained with $h(p) = -p \log(p)$ and $h(p) = (\alpha - 1)^{-1} (p - p^\alpha)$, $\alpha > 1$, corresponding respectively to the classical Shannon (or Boltzmann-Gibbs) entropy, $H(\nu) = -\sum_y \nu_y \log(\nu_y)$, and to the α -entropy of [Havrda and Charvát \(1967\)](#) and [Tsallis \(1988\)](#), $H(\nu) = (\alpha - 1)^{-1} (1 - \sum_y \nu_y^\alpha)$. The special case $\alpha = 2$ corresponds to the Gini-Simpson diversity index ([Simpson, 1949](#)), $H(\nu) = 1 - \sum_y \nu_y^2$.

2.2.2 Regular quadratic uncertainty functionals

For any symmetric measurable function $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$, the expression

$$H(\nu) = \iint \rho(y, y') \nu(dy) \nu(dy') \quad (7)$$

defines a measurable probability functional $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}_+$. Uncertainty functionals of this form have been studied, for example, by [Rao \(1984\)](#), under the name *quadratic entropy*, and by [Hainy et al. \(2014\)](#), who calls *distance-based information function* the negative of such an uncertainty functional. More generally, functionals of the form (7)—not necessarily uncertainty functionals—are called *regular statistical functionals of degree 2* in the literature on U-statistics (see, e.g., [Lee, 1990](#)).

Remark 6. In the terminology of potential theory, $H(\nu)$ corresponds to the negative of the *energy* of ν (see, e.g., [Pronzato and Zhigljavsky, 2021](#)) associated with the kernel ρ .

Note, however, that classical potential theory typically deals with singular kernels that are infinite on the diagonal, such as the Newtonian kernel $\rho(x, x') \propto \|x - x'\|^{-1}$ on $\mathbb{R}^3$. In contrast, we shall be mainly interested in kernels that vanish on the diagonal (see [Theorem 1](#)).

Definition 3. Any uncertainty functional of the form (7) is called a *regular quadratic uncertainty functional* (regular QUF). The symmetric measurable function $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \overline{\mathbb{R}}_+$ is called the *kernel* of the regular QUF.

Let $\mathbb{P}_H = \{\nu \in \mathbb{P} \mid H(\nu) < +\infty\}$, and $\mathbb{M}_H = \text{span}(\mathbb{P}_H)$ denote the vector space of signed measures spanned by $\mathbb{P}_H$. The following result provides a characterization of regular QUFs in terms of properties of their kernel.

Theorem 1. Let $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}_+$ denote a measurable, non-negative functional of the form (7), where $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$ is symmetric and measurable. Then, H is a regular QUF if, and only if,

- i) ρ vanishes on the diagonal (i.e., $\rho(y, y) = 0$ for all $y \in \mathbb{Y}$), and
- ii) ρ is conditionally integrally negative definite (CIND), i.e.,

$$\forall \nu \in \mathbb{M}_H, \quad \nu(\mathbb{Y}) = 0 \Rightarrow \iint \rho(y, y') \nu(dy) \nu(dy') \leq 0. \quad (8)$$

Moreover, if H is a regular QUF, then $\mathbb{P}_H$ is convex, and for all random probability measures ν on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$ such that $H(\nu) < +\infty$, we have

$$H(\bar{\nu}) - \mathbb{E}(H(\nu)) = -\mathbb{E} \left[\iint \rho(y, y') (\nu - \bar{\nu})(dy) (\nu - \bar{\nu})(dy') \right]. \quad (9)$$

Proof. See [Section A.1](#) for a proof. Note: [Pronzato and Zhigljavsky \(2020, Lemma 3.2\)](#) proved a similar result under the stronger assumption that the kernel is bounded. The proof presented in [Section A.1](#) has a lot in common with that of [Pronzato and Zhigljavsky](#), but it further generalizes the argument to cover unbounded kernels as well. $\square$

Several important uncertainty functionals can be recovered in this framework, including the variance functional $H(\nu) = \int y^2 \nu(dy) - (\int y \nu(dy))^2$ on $\mathbb{Y} = \mathbb{R}$ with $\rho(y, y') = \frac{1}{2}(y - y')^2$, Gini's mean difference ([Yitzhaki, 2003](#)) with $\rho(y, y') = |y - y'|$, or more generally the fractional Brownian variance of [Székely and Rizzo \(2013\)](#) with $\rho(y, y') = \|y - y'\|^\beta$, $0 < \beta < 2$, on $\mathbb{Y} = \mathbb{R}^p$. The Gini-Simpson index $H(\nu) = 1 - \sum_y \nu_y^2$ on a finite or countable set $\mathbb{Y}$, already discussed in [Section 2.2.1](#), is also recovered here with $\rho(y, y') = \mathbb{1}_{y \neq y'}$.

Additionally, given a σ -finite measure space $(\mathbb{T}, \mathcal{T}, \mu)$ and a measurable function $\xi : \mathbb{Y} \times \mathbb{T} \rightarrow \mathbb{R}$, the kernel

$$\rho(y, y') = \frac{1}{2} \int_{\mathbb{T}} (\xi(y, t) - \xi(y', t))^2 \mu(dt)$$

yields the integrated variance functional:

$$H(\nu) = \int_{\mathbb{T}} \text{var}_{\nu}(\xi(Y, t)) \mu(dt), \quad (10)$$

where the index ν indicates that $Y \sim \nu$. The Gini-Simpson index and the integrated Bernoulli variance functional (Bect et al., 2019) are both special cases of Eq. (10), where ξ takes its values in $\{0, 1\}$.

The term CIND, introduced in Theorem 1, does not appear to have been used previously in the literature. It is, however, consistent with the terminology used by Pronzato and Zhigljavsky (2023), since ρ is CIND in the sense of Theorem 1 if and only if it is *conditionally integrally positive definite* (CIPD) in the sense of Pronzato and Zhigljavsky (2023). See also Pronzato and Zhigljavsky (2020, 2021) and Sriperumbudur et al. (2011) for related terminology.

Proposition 3. *Let $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$ be symmetric, measurable and vanishing on the diagonal. Then ρ is CIND if, and only if, it is conditionally negative definite (CND), that is, for all $m \geq 1$, all points $y_1, \dots, y_m \in \mathbb{Y}$, and all $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ such that $\sum_{k=1}^m \lambda_k = 0$,*

$$\sum_{k,l}^m \lambda_k \lambda_l \rho(y_k, y_l) \leq 0. \quad (11)$$

Remark 7. It follows from classical results on CND functions that ρ is symmetric, CND and vanishes on the diagonal if, and only, there exists a symmetric positive definite kernel $k : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}$ such that $\rho(y, y') = \|k(y, \cdot) - k(y', \cdot)\|_{\mathbb{H}}^2$, where $\mathbb{H}$ is the reproducing kernel Hilbert space with kernel k (see Theorem 7 and Remark 15 for details). In this setting, for all $\nu \in \mathbb{P}_H$, the kernel embedding $k_{\nu} = \int k(y, \cdot) \nu(dy)$ is well-defined (see Lemma 5 and Remark 16). Moreover, for all random probability measures ν on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$ such that $\bar{\nu} \in \mathbb{P}_H$, we have $\nu \in \mathbb{P}_H$ almost surely, and it is easily checked that the right-hand side of Eq. (9) is equal to $\mathbb{E}(\|k_{\nu} - k_{\bar{\nu}}\|_{\mathbb{H}}^2)$. In other words, the (unnormalized) sensitivity indices based on a regular QUF coincide with the MMD-based indices of Da Veiga (2021).

3 Concavity and Jensen's inequality

3.1 Main results and an open problem

We turn now to the problem of characterizing the probability functionals that satisfy Jensen's inequality. More specifically, we investigate the relation between Jensen's inequality and concavity. This is indeed a natural question, considering the well-known equivalence, for functions defined on $\mathbb{R}^n$, between concavity and Jensen's inequality. We recall here this equivalence, in form that is suitable for comparison with Definition 1.

Theorem 2 (Jensen’s inequality on $\mathbb{R}^n$). *Let $f : \mathbb{R}^n \rightarrow \overline{\mathbb{R}}$ be a measurable function. Then, f is concave if and only if, for all random vectors X in $\mathbb{R}^n$ such that $\mathbb{E}(\|X\|) < +\infty$ and $f(X)$ is quasi-integrable, the inequality $f(\mathbb{E}(X)) \geq \mathbb{E}(f(X))$ holds.*

Remark 8. For concave functions with values in $\mathbb{R} \cup \{-\infty\}$, the statement can be strengthened: in this case, $\mathbb{E}(f^+(X)) < +\infty$, and thus $f(X)$ is automatically quasi-integrable.

Concavity for probability functionals is defined as follows. (This is the usual definition of concavity for extended-real valued functions, when $\mathbb{P}$ is seen as a convex subset of the set of bounded signed measures on $(\mathbb{Y}, \mathcal{Y})$.)

Definition 4. A probability functional $H : \mathbb{P} \rightarrow \overline{\mathbb{R}}$ is called *concave* if, for all $\lambda \in (0, 1)$ and all $\nu_1, \nu_2 \in \text{dom}(H) = \{\nu \in \mathbb{P} \mid H(\nu) > -\infty\}$,

$$H(\lambda\nu_1 + (1 - \lambda)\nu_2) \geq \lambda H(\nu_1) + (1 - \lambda)H(\nu_2). \quad (12)$$

In the case where $\mathbb{Y}$ is a finite set, DeGroot (1962, Theorem 2.1) established, for probability functionals, the equivalence between concavity and the property of being decreasing on average (which is equivalent, by Proposition 1, to Jensen’s inequality). In fact, a part of DeGroot’s argument remains valid for a general measurable space $(\mathbb{Y}, \mathcal{Y})$, which yields the following statement.

Theorem 3 (DeGroot, 1962). *Let H denote a probability functional on a measurable space $(\mathbb{Y}, \mathcal{Y})$. If H satisfies Jensen’s inequality, then H is concave. Conversely, if H is concave and $\mathbb{Y}$ is finite, then H satisfies Jensen’s inequality.*

Proof. We follow DeGroot’s original proof, with minor adaptations.

Let H be a probability functional that satisfies Jensen’s inequality. Let $\lambda \in (0, 1)$ and $\nu, \nu' \in \text{dom}(H)$. Let $\bar{\nu}$ be a random measure equal to ν with probability λ , and ν' otherwise. Then $\bar{\nu} = \lambda\nu + (1 - \lambda)\nu'$, and $H(\bar{\nu})$ is quasi-integrable (since $\mathbb{E}(H^-(\bar{\nu})) < +\infty$). Applying Jensen’s inequality to $\bar{\nu}$ yields the concavity of H .

Conversely, assume that $\mathbb{Y}$ has finite cardinality n , and let H be a concave probability functional on $(\mathbb{Y}, \mathcal{Y})$ for some σ -algebra $\mathcal{Y}$ on $\mathbb{Y}$. Then the set $\mathbb{P}$ of all probability measures on $(\mathbb{Y}, \mathcal{Y})$ can be identified measurably with a convex subset C of the $(n - 1)$ -dimensional probability simplex in $\mathbb{R}^n$. Thus, H can be identified with a concave function on $\mathbb{R}^n$ (using the value $-\infty$ to extend outside C), and therefore Jensen’s inequality for the probability functional H follows from Theorem 2. (Note: $\mathbb{P}$ coincides exactly with the probability simplex if $\mathcal{Y} = \mathcal{P}(\mathbb{Y})$.) $\square$

Hainy et al. (2014, Theorem 2) states a generalization of Theorem 3, formulated using the property of being decreasing on average (recall Proposition 1) rather than Jensen’s inequality (Definition 1), which removes the finiteness restriction on $\mathbb{Y}$ in the converse. The proposed proof, however, is not sufficient: in particular, it does not specify how the use of Theorem 2 can be replaced, since in general $\mathbb{P}$ can no longer be identified with a convex subset of some Euclidean space, when $\mathbb{Y}$ is not finite.

We are now in a position to state our main results, which prove that the equivalence between concavity and Jensen’s inequality does not hold for probability functionals

on general measurable spaces. Our first result is based on an earlier counterexample, proposed by [Perlman \(1974\)](#) in the setting of convex functionals defined on topological vector spaces. Our second result relies on a different type of counterexample, which is new to the best of our knowledge.

Theorem 4. *Assume that $(\mathbb{Y}, \mathcal{Y}) \cong (\mathbb{N}, \mathcal{P}(\mathbb{N})) \otimes (\mathbb{Y}', \mathcal{Y}')$, for some measurable space $(\mathbb{Y}', \mathcal{Y}')$. Then, there exists a concave, measurable probability functional on $(\mathbb{Y}, \mathcal{Y})$ that does not satisfy Jensen's inequality.*

Theorem 5. *Assume that $(\mathbb{Y}, \mathcal{Y}) \cong (\mathbb{R}, \mathcal{B}(\mathbb{R})) \otimes (\mathbb{Y}', \mathcal{Y}')$, for some measurable space $(\mathbb{Y}', \mathcal{Y}')$. Then, there exists a concave, non-negative, measurable probability functional on $(\mathbb{Y}, \mathcal{Y})$ that vanishes on Dirac measures but does not satisfy Jensen's inequality.*

Proof. For both results we can assume without loss of generality that $(\mathbb{Y}', \mathcal{Y}')$ is trivial, i.e., that $(\mathbb{Y}, \mathcal{Y}) \cong (\mathbb{N}, \mathcal{P}(\mathbb{N}))$ for [Theorem 4](#) and $(\mathbb{Y}, \mathcal{Y}) \cong (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for [Theorem 5](#). A concave probability functional that does not satisfy Jensen's inequality is constructed, for each of the two settings, in [Sections 3.2.1](#) and [3.2.2](#) respectively. $\square$

[Theorem 5](#) proves that the requirement that H should satisfy Jensen's inequality, in the definition of an uncertainty functional, cannot be replaced by a simple concavity requirement as soon as the underlying measurable space $(\mathbb{Y}, \mathcal{Y})$ “carries” a real-valued random variable (in the sense that $(\mathbb{Y}, \mathcal{Y}) \cong (\mathbb{R}, \mathcal{B}(\mathbb{R})) \otimes (\mathbb{Y}', \mathcal{Y}')$, for some measurable space $(\mathbb{Y}', \mathcal{Y}')$). In the case of uncertainty functionals on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$, the question remains open.

Open Problem 1. *Is it possible to construct a concave, non-negative, measurable probability functional on $(\mathbb{N}, \mathcal{P}(\mathbb{N}))$ that vanishes on Dirac measures but does not satisfy Jensen's inequality?*

3.2 Counterexamples for Jensen's inequality

3.2.1 A first counterexample (proof of [Theorem 4](#))

In this section, we present a counterexample adapted from [Perlman \(1974\)](#) as a proof of [Theorem 4](#). Take $\mathbb{Y} = \mathbb{N}$, and let H denote the real-valued probability functional defined by

$$H(\nu) = \begin{cases} g(\nu) & \text{if } g(\nu) > -\infty, \\ 0 & \text{if } g(\nu) = -\infty, \end{cases} \quad \text{where } g(\nu) = \liminf_{y \rightarrow +\infty} (-y^2 \nu_y). \quad (13)$$

Note that H takes values in $(-\infty, 0]$.

We begin by establishing concavity. For any $\nu, \nu' \in \mathbb{P}$ and $\lambda \in (0, 1)$, if $g(\lambda\nu + (1-\lambda)\nu') = -\infty$, we have $H(\lambda\nu + (1-\lambda)\nu') = 0$ which reaches the maximum of H , and thus clearly satisfies the desired inequality. In the case where the

$g(\lambda\nu + (1 - \lambda)\nu')$ is finite, we have

$$\begin{aligned} -\infty < H(\lambda\nu + (1 - \lambda)\nu') &= \liminf_{y \rightarrow +\infty} [-y^2 (\lambda\nu_y + (1 - \lambda)\nu'_y)] \\ &\leq \min \left\{ \lambda \liminf_{y \rightarrow +\infty} (-y^2 \nu_y), (1 - \lambda) \liminf_{y \rightarrow +\infty} (-y^2 \nu'_y) \right\}, \end{aligned}$$

which implies that $g(\nu)$ and $g(\nu')$ are both finite, indicating that we have both $H(\nu) = g(\nu)$ and $H(\nu') = g(\nu')$. Hence, the superadditivity and positive homogeneity of the $\liminf$ operator imply that

$$\begin{aligned} H(\lambda\nu + (1 - \lambda)\nu') &= g(\lambda\nu + (1 - \lambda)\nu') \\ &\geq \lambda g(\nu) + (1 - \lambda)g(\nu') = \lambda H(\nu) + (1 - \lambda)H(\nu'). \end{aligned}$$

To see that Jensen's inequality does not hold for this functional, consider a discrete random variable X taking values in $\mathbb{N}$, with distribution

$$\mathbb{P}(X = x) = \frac{6}{\pi^2} \frac{1}{(1 + x)^2}, \quad x \in \mathbb{N},$$

and the random measure $\nu = \delta_X$, where δ_x denotes the Dirac measure at x . In this setting we have

$$H(\bar{\nu}) = H(\mathbb{P}^X) = \frac{6}{\pi^2} \liminf_{y \rightarrow +\infty} \left(-\frac{y^2}{(1 + y)^2} \right) = -\frac{6}{\pi^2},$$

and $H(\nu) = 0$ since H vanishes on Dirac measures. Therefore $H(\bar{\nu}) < \mathbb{E}(H(\nu))$, which contradicts Jensen's inequality.

Remark 9. In fact, the function g alone is sufficient to show [Theorem 4](#). Its range covers the extended non-positive real line, whereas the range of H is essentially the same, except that it excludes $-\infty$. This construction is intended to highlight that the issue does not stem from the infinite value.

3.2.2 Another counterexample (proof of [Theorem 5](#))

In this section, we construct a concave probability functional that satisfies all the requirements of an uncertainty functional except Jensen's inequality, thereby proving [Theorem 5](#).

Let $\mathbb{Y} = \mathbb{R}$ and $\mathcal{Y} = \mathcal{B}(\mathbb{R})$. Consider the integrated variance functional defined by

$$H(\nu) = \int_{\mathbb{R}} \text{var}_{\nu}(\mathbb{1}_{Y=s}) \mu(ds), \quad (14)$$

where μ denotes the counting measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The probability functional H is measurable, nonnegative, and vanishes on Dirac measures. It is also concave, since for

each $s \in \mathbb{R}$ the mapping $\nu \mapsto \text{var}_\nu(\mathbb{1}_{Y=s})$ is concave (and in fact satisfies Jensen's inequality).

To see that Jensen's inequality does not hold for H , observe that

$$H(\nu) = \int_{\mathbb{R}} \nu_s (1 - \nu_s) \mu(ds) = \sum_{s \in \mathbb{R}} \nu_s (1 - \nu_s), \quad (15)$$

where $\nu_s = \nu(\{s\})$ as before. Consider the random probability measure $\boldsymbol{\nu} = \frac{1}{2}(\delta_{-X} + \delta_X)$, where X follows an exponential distribution. Then $\bar{\nu}$ is a Laplace distribution, which has no atom and hence $H(\bar{\nu}) = 0$. On the other hand, $H(\boldsymbol{\nu}) = \frac{1}{2}$ almost surely. Therefore Jensen's inequality does not hold:

$$\mathbb{E}(H(\boldsymbol{\nu})) = \frac{1}{2} > 0 = H(\bar{\nu}).$$

Remark 10. The essence of this counterexample is the lack of σ -finiteness of the counting measure μ on $\mathbb{R}$. Indeed, as mentioned in [Section 2.2.2](#), any probability functional of the form (14) with μ a σ -finite measure is an uncertainty functional, which is an easy consequence of Fubini–Tonelli's theorem:

$$\begin{aligned} H(\bar{\nu}) &= \int_{\mathbb{R}} \text{var}_{\bar{\nu}}(\mathbb{1}_{Y=s}) \mu(ds) \geq \int_{\mathbb{R}} \mathbb{E}(\text{var}_{\boldsymbol{\nu}}(\mathbb{1}_{Y=s})) \mu(ds) \\ &= \mathbb{E}\left(\int_{\mathbb{R}} \text{var}_{\boldsymbol{\nu}}(\mathbb{1}_{Y=s}) \mu(ds)\right) = \mathbb{E}(H(\boldsymbol{\nu})). \end{aligned}$$

Remark 11. It is easy to see from Eq. (15) that H is upper-bounded by one. More precisely, $\sup H = 1$ and the sup is not attained.

3.3 A sufficient condition, and more open problems

We focus in this section on non-negative probability functionals. A sufficient condition for Jensen's inequality to hold in this setting is provided by the following result.

Proposition 4 (see, e.g., [Bect et al., 2019](#), Proposition 3.17). *Let $\mathbb{D}$ be a non-empty set. Let $L : \mathbb{Y} \times \mathbb{D} \rightarrow \bar{\mathbb{R}}_+$ be a function such that, for all $d \in \mathbb{D}$, $L(\cdot, d)$ is measurable. Then, the expression*

$$H(\nu) = \inf_{d \in \mathbb{D}} \int L(y, d) \nu(dy), \quad \nu \in \mathbb{P}, \quad (16)$$

defines a non-negative measurable probability functional $H : \mathbb{P} \rightarrow \bar{\mathbb{R}}_+$ that satisfies Jensen's inequality.

Remark 12. The probability function (16) is an uncertainty functional in the sense of [Definition 2](#) if, and only if, $\inf L(y, \cdot) = 0$ for all $y \in \mathbb{Y}$.

Remark 13. Sensitivity indices based on probability functionals of the form (16) have been proposed by Fort et al. (2016), Borgonovo et al. (2021), Fissler and Pesenti (2023) and Straub et al. (2025).

Interpreting $d \in \mathbb{D}$ as a decision, L as a loss function, and ν as a posterior distribution in a Bayesian setting, the integral in the right-hand-side of Eq. (16) corresponds to the posterior risk, and the uncertainty $H(\nu)$ to the minimal posterior risk. This idea of measuring uncertainty using an expected loss—or information using an expected utility—has a long history, which goes back to the early days of statistical decision theory (see, e.g., McCarthy, 1956; DeGroot, 1962, in particular Eq. (2.14) and the associated discussion). See also Bernardo (1979), DeGroot (1986, 1994), Dawid (2007) and Ginebra (2007). The following question, however, remains to the best of our knowledge unanswered (see below for a partial answer).

Open Problem 2. *Does there exist a measurable space $(\mathbb{Y}, \mathcal{Y})$, and a non-negative measurable probability functional H on $(\mathbb{Y}, \mathcal{Y})$, such that H satisfies Jensen’s inequality but cannot be written in the form (16)?*

As a special case, consider now the situation where the infimum in Eq. (16) is attained for all $\nu \in \mathbb{P}$. It turns out that this corresponds to a well-known class of uncertainty functionals (see, e.g., Grünwald and Dawid, 2004; Dawid, 2007; Gneiting and Raftery, 2007):

Proposition 5. *A probability functional H admits the representation (16), where the infimum on $\mathbb{D}$ is attained for all $\nu \in \mathbb{P}$ if, and only if, it is the generalized entropy function associated with a proper, non-negative, negatively-oriented scoring rule $S : \mathbb{P} \times \mathbb{Y} \rightarrow \bar{\mathbb{R}}_+$.*

Proof. We reproduce the argument from Grünwald and Dawid (2004, Section 3.4).

Assume that H has the form (16), and the infimum is attained for all $\nu \in \mathbb{P}$. Then, there is a function $\nu \mapsto d_\nu^*$ from $\mathbb{P}$ to $\mathbb{D}$ such that, for all $\nu \in \mathbb{P}$,

$$H(\nu) = \int L(y, d_\nu^*) \nu(dy) = \min_{\nu' \in \mathbb{P}} \int L(y, d_{\nu'}^*) \nu(dy). \quad (17)$$

Setting $S(y, \nu) = L(y, d_\nu^*)$ yields the desired negatively-oriented scoring rule, which takes values in $\bar{\mathbb{R}}_+$, and is proper since the minimum in Eq. (17) is attained at $\nu' = \nu$.

Conversely, assume that there exists a proper, non-negative, negatively-oriented scoring rule $S : \mathbb{P} \times \mathbb{Y} \rightarrow \bar{\mathbb{R}}_+$ such that, for all $\nu \in \mathbb{P}$,

$$H(\nu) = \int S(y, \nu) \nu(dy) = \min_{\nu' \in \mathbb{P}} \int S(y, \nu') \nu(dy).$$

Then the representation (16) holds with $\mathbb{D} = \mathbb{P}$ and $L = S$. $\square$

Open Problem 3. *Does there exist a measurable space $(\mathbb{Y}, \mathcal{Y})$, and a non-negative measurable probability functional H on $(\mathbb{Y}, \mathcal{Y})$, such that H can be written in the form (16), but does not have an associated proper scoring rule?*

We conclude this section with a result that provides a partial answer to Open Problems 2 and 3: the answer to both questions is negative in the case of a finite set $\mathbb{Y}$.

Theorem 6. *Let $\mathbb{Y}$ be a finite set, endowed with a σ -algebra $\mathcal{Y}$. Let H denote a non-negative, concave probability functional on $(\mathbb{Y}, \mathcal{Y})$. Then, H can be written in the form (16), in such a way that the infimum is attained for all $\nu \in \mathbb{P}$.*

Proof. See Section A.3. □

Remark 14. It is critical, for this result to hold, to allow the value $+\infty$ for the function L in (16). A special case of this result is proved by DeGroot (1962), under the additional assumption that H , seen as a function on the probability simplex, is continuous. In this case, L can be chosen to take only finite values, and the result is a straightforward consequence of a classical result in convex analysis—namely, that a continuous concave function on a convex subset of $\mathbb{R}^n$ admits at each point an exact affine majorant.

4 Conclusion

In this article, we have provided a unified theoretical treatment of uncertainty functionals on general measurable spaces, going beyond DeGroot’s seminal work on finite spaces. We have proved that concavity is not, in general, a sufficient condition for Jensen’s inequality to hold in its probabilistic form—a key property in applications such as Bayesian experimental design and global sensitivity analysis. In particular, our second counterexample demonstrates that Jensen’s inequality can fail to hold even for non-negative bounded concave functionals that vanish on Dirac measures.

Several important questions remain open. In particular, to further clarify the theory of uncertainty functionals, it is important to understand whether all uncertainty functionals arise as the generalized entropy function of some proper scoring rule. Another important direction for future research is to refine the analysis of uncertainty functionals in the context of global sensitivity analysis. Indeed, the property of being decreasing on average is necessary but not sufficient to yield a complete set of sensitivity indices à la Sobol’, in which interactions are separated from main effects in a meaningful way.

Appendix A Proofs

A.1 Proof of Theorem 1

Theorem 7 (see, e.g., Berg et al., 1984, Chapter 3). *Let $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}$ be symmetric and vanishing on the diagonal. Then, ρ is CND if, and only if, there exists a Hilbert space $\mathbb{H}$ and a function $\psi : \mathbb{Y} \rightarrow \mathbb{H}$, such that $\rho(y, y') = \|\psi(y) - \psi(y')\|^2$ for all $y, y' \in \mathbb{Y}$.*

We will consider in Appendices A.1 and A.2 a measurable kernel ρ , a Hilbert space $\mathbb{H}$ and a function $\psi : \mathbb{Y} \rightarrow \mathbb{H}$ with the same properties as in Theorem 7, which we summarize for reference:

Assumption 1. $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}$ is a symmetric and measurable CND kernel, vanishing on the diagonal, and $\psi : \mathbb{Y} \rightarrow \mathbb{H}$ is such that $\rho(y, y') = \|\psi(y) - \psi(y')\|^2$ for all $y, y' \in \mathbb{Y}$.

Remark 15. Given a kernel ρ that satisfies [Assumption 1](#), a corresponding function $\psi : \mathbb{Y} \rightarrow \mathbb{H}$ for ρ is obtained, for instance, by considering a reproducing kernel $\mathbb{H}$ Hilbert space on $\mathbb{Y}$, with symmetric positive definite kernel k such that $\rho(y, y') = k(y, y) + k(y', y') - 2k(y, y')$, and setting $\psi(y) = k(y, \cdot)$. It is well known that such a kernel k always exists if ρ is symmetric, CND and vanishes on the diagonal: for instance, take $k(y, y') = \frac{1}{2}(\rho(y, y_0) + \rho(y', y_0) - \rho(y, y'))$ for some arbitrary $y_0 \in \mathbb{Y}$. (Note that k is measurable.)

Lemma 1. *Let [Assumption 1](#) hold. Then ψ is weakly measurable, i.e., $y \mapsto \langle h, \psi(y) \rangle$ is measurable for all $h \in \mathbb{H}$.*

Proof. Assume first that $\mathbb{H}$ and ψ are as in [Remark 15](#). Then, for all $h \in \mathbb{H}$ of the form $h = k(y_0, \cdot)$, we have $\langle h, \psi(y) \rangle = k(y_0, y)$, and therefore $y \mapsto \langle h, \psi(y) \rangle$ is measurable since k is measurable. The result extends to linear combinations, and then to all $h \in \mathbb{H}$ by density. Therefore, ψ is weakly measurable.

Assume now that $\tilde{\psi} : \mathbb{Y} \rightarrow \tilde{\mathbb{H}}$ provides another representation of the same kernel ρ . Then there exists an isometric isomorphism between the closure of $\text{span}\{\psi(y) - \psi(y_0), y \in \mathbb{Y}\}$ in $\mathbb{H}$ and the closure of $\text{span}\{\tilde{\psi}(y) - \tilde{\psi}(y_0), y \in \mathbb{Y}\}$ in $\tilde{\mathbb{H}}$, from which it is easy to see that $\tilde{\psi}$ is weakly measurable. $\square$

Lemma 2. *Assume that [Assumption 1](#) holds, and let $\nu \in \mathbb{P}$. Then $\iint \rho d(\nu \otimes \nu) < +\infty$ if, and only if, $\int \|\psi\|^2 d\nu < +\infty$. As a consequence, the set $\mathbb{P}_H = \{\nu \in \mathbb{P} \mid \iint \rho d(\nu \otimes \nu) < +\infty\}$ is convex.*

Proof. Preliminary remark: since ψ is weakly measurable by [Lemma 1](#), $y \mapsto \|\psi(y)\|^2$ is measurable as well, since $\|\psi(y)\|^2 = \sup_{\|h\| \leq 1} \langle h, \psi(y) \rangle$.

Let $\nu \in \mathbb{P}$ be such that $\int \|\psi\|^2 d\nu < +\infty$. Then we have $\rho(y, y') \leq 2(\|\psi(y)\|^2 + \|\psi(y')\|^2)$ and thus

$$\iint \rho d(\nu \otimes \nu) \leq 4 \int \|\psi\|^2 d\nu < +\infty.$$

Conversely, assume that $\iint \rho d(\nu \otimes \nu) < +\infty$. Then, for ν -almost all $y_0 \in \mathbb{Y}$, we have $\int \rho(y, y_0) \nu(dy) < +\infty$. For any $y_0 \in \mathbb{Y}$ such that this inequality holds, we get

$$\begin{aligned} \int \|\psi\|^2 d\nu &\leq 2 \int \left[\|\psi(y) - \psi(y_0)\|^2 + \|\psi(y_0)\|^2 \right] \nu(dy) \\ &= 2 \int \rho(y, y_0) \nu(dy) + 2 \|\psi(y_0)\|^2 < +\infty. \end{aligned} \quad \square$$

Lemma 3. *Let [Assumption 1](#) hold. Then, for all $\nu_1, \nu_2 \in \mathbb{P}_H$, $\rho \in L^1(\nu_1 \otimes \nu_2)$.*

Proof. Let $\nu_1, \nu_2 \in \mathbb{P}_H$. Set

$$B(\nu_1, \nu_2) = \iint_{\mathbb{Y} \times \mathbb{Y}} \rho(y, y') \nu_1(dy) \nu_2(dy') \in [0, +\infty],$$

and note that $H(\nu) = B(\nu, \nu)$. The set $\mathbb{P}_H$ is convex by [Lemma 2](#), and therefore $H(\frac{1}{2}(\nu_1 + \nu_2)) < +\infty$ for all $\nu_1, \nu_2 \in \mathbb{P}_H$. Besides, B is symmetric since ρ is symmetric. Thus

$$\begin{aligned} H\left(\frac{\nu_1 + \nu_2}{2}\right) &= \iint \rho d\left[\frac{1}{4}\nu_1 \otimes \nu_1 + \frac{1}{4}\nu_1 \otimes \nu_2 + \frac{1}{4}\nu_2 \otimes \nu_1 + \frac{1}{4}\nu_2 \otimes \nu_2\right] \\ &= \frac{1}{4}H(\nu_1) + \frac{1}{4}H(\nu_2) + \frac{1}{2}B(\nu_1, \nu_2), \end{aligned}$$

which proves that $B(\nu_1, \nu_2)$ is finite. $\square$

Proof of Theorem 1. To begin with, observe that $H(\delta_y) = \rho(y, y)$ for all $y \in \mathbb{Y}$. Thus, it is clear that H vanishes on Dirac measures if, and only if, its kernel ρ vanishes on the diagonal. We assume from now on that $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$ is symmetric, measurable, and vanishes on the diagonal.

Assume that H is an uncertainty functional, and therefore satisfies Jensen's inequality. Then, in particular, H is concave by [Theorem 3](#), which implies that ρ is conditionally negative definite (CND). To see it, set $\nu = \sum_{k=1}^m \lambda_k \delta_{y_k}$. Assume without loss of generality that $\nu \neq 0$. Then there exists finitely supported probability measures $\nu_1, \nu_2 \in \mathbb{P}_H$, and $\alpha > 0$, such that $\nu = \alpha(\nu_1 - \nu_2)$. Setting $\bar{\nu} = \frac{1}{2}(\nu_1 + \nu_2)$, it is easy to check that

$$\sum_{k,l} \lambda_k \lambda_l \rho(y_k, y_l) = \iint \rho d(\nu \otimes \nu) = 4\alpha^2 \left(\frac{1}{2}(H(\nu_1) + H(\nu_2)) - H(\bar{\nu}) \right) \leq 0,$$

which proves that ρ is CND. Now let $\nu \in \mathbb{M}_H$ be such that $\nu(\mathbb{Y}) = 0$. Again, assume without loss of generality that $\nu \neq 0$, since Eq. (8) trivially holds for $\nu = 0$. Then there exists $\nu_1, \nu_2 \in \mathbb{P}_H$, and $\alpha > 0$, such that $\nu = \alpha(\nu_1 - \nu_2)$. We know from [Lemma 3](#) that $\rho \in L^1(\nu_1 \otimes \nu_2)$, and therefore $\rho \in L^1(|\nu \otimes \nu|)$. Finally, setting $\bar{\nu} = \frac{1}{2}(\nu_1 + \nu_2)$, we have

$$\iint \rho d(\nu \otimes \nu) = 4\alpha^2 \left(\frac{1}{2}(H(\nu_1) + H(\nu_2)) - H(\bar{\nu}) \right) \leq 0,$$

which proves that ρ is CIND.

Conversely, assume that ρ is CIND, and therefore CND. Let ν be a random probability measures on $(\mathbb{Y}, \mathcal{P}(\mathbb{Y}))$. Assume without loss of generality that $H(\bar{\nu}) < +\infty$, i.e., $\bar{\nu} \in \mathbb{P}_H$ (if not, Jensen's inequality holds trivially). Then, by [Lemma 2](#), we have $\int \|\psi\|^2 d\bar{\nu} < +\infty$. Since $\int \|\psi\|^2 d\bar{\nu} = \mathbb{E}(\int \|\psi\|^2 d\nu)$, we also have $\int \|\psi\|^2 d\nu < +\infty$ almost surely, hence $\nu \in \mathbb{P}_H$ almost surely. The integral $\iint \rho d[(\nu - \bar{\nu}) \otimes (\nu - \bar{\nu})]$ in the right-hand side of Eq. (9) is therefore well-defined almost surely. Using the bilinearity of $\otimes$ and the symmetry of ρ , we have:

$$-\iint \rho d[(\nu - \bar{\nu}) \otimes (\nu - \bar{\nu})] = 2 \iint \rho d(\nu \otimes \bar{\nu}) - H(\nu) - H(\bar{\nu}).$$

Finally, taking expectations we obtain

$$-\mathbb{E} \left(\iint \rho d[(\nu - \bar{\nu}) \otimes (\nu - \bar{\nu})] \right) = 2H(\bar{\nu}) - \mathbb{E}(H(\nu)) - H(\bar{\nu}) = H(\bar{\nu}) - \mathbb{E}(H(\nu)), \quad (\text{A1})$$

where we have used Fubini–Tonelli's theorem and the fact that $\mathbb{E}(\int \rho(\cdot, y') d\nu) = \int \rho(\cdot, y') d\bar{\nu}$. The left-hand side is non-negative since ρ is CIND, $\nu - \bar{\nu} \in \mathbb{M}_H$ almost surely and $(\nu - \bar{\nu})(\mathbb{Y}) = 0$, which concludes the proof. $\square$

A.2 Proof of Proposition 3

Lemma 4. *Let Assumption 1 hold. Let $\mathbb{M}_H$ be defined as in Section 2.2.2, i.e., $\mathbb{M}_H = \text{span}(\mathbb{P}_H)$, with $\mathbb{P}_H = \{\nu \in \mathbb{P} \mid \iint \rho d(\nu \otimes \nu) < +\infty\}$. Then we have*

$$\mathbb{M}_H = \{\nu \in \mathbb{M} \mid \iint \rho d|\nu \otimes \nu| < +\infty\} = \{\nu \in \mathbb{M} \mid \int \|\psi\|^2 d|\nu| < +\infty\}, \quad (\text{A2})$$

where $\mathbb{M}$ denotes the space of bounded signed measures on $\mathbb{Y}$.

Proof. By Lemma 2 we have the equivalence

$$\iint \rho d(\nu \otimes \nu) < +\infty \Leftrightarrow \int \|\psi\|^2 d\nu < +\infty \quad (\text{A3})$$

for all $\nu \in \mathbb{P}$, which extends immediately to non-negative bounded measures.

Let $\nu \in \mathbb{M}_H = \text{span}(\mathbb{P}_H)$. Then there exists $\nu_1, \nu_2 \in \mathbb{P}_H$ and $\alpha, \beta \geq 0$ such that $\nu = \alpha\nu_1 - \beta\nu_2$. By minimality of the Hahn-Jordan decomposition, it follows that

$$\int \|\psi\|^2 d|\nu| = \int \|\psi\|^2 d\nu^+ + \int \|\psi\|^2 d\nu^- \leq \alpha \int \|\psi\|^2 d\nu_1 + \beta \int \|\psi\|^2 d\nu_2 < +\infty.$$

Conversely, any $\nu \in \mathbb{M}$ such that $\int \|\psi\|^2 d|\nu| < +\infty$ can be written using the Hahn-Jordan decomposition as $\nu = \alpha\nu_1 - \beta\nu_2$, where ν_1 and ν_2 are probability measures such that $\int \|\psi\|^2 d\nu_j < +\infty$, $j \in \{1, 2\}$, which implies that $\nu_1, \nu_2 \in \mathbb{P}_H$, and therefore $\nu \in \text{span}(\mathbb{P}_H)$.

So far we have proved that $\mathbb{M}_H = \{\nu \in \mathbb{M} \mid \int \|\psi\|^2 d|\nu| < +\infty\}$. The other part of Eq. (A2) then follows from Eq. (A3), using the fact that $|\nu \otimes \nu| = |\nu| \otimes |\nu|$. $\square$

Lemma 5. *Let Assumption 1 hold, and let $\nu \in \mathbb{M}_H$. Then ψ is Pettis-integrable with respect to ν , i.e., there exists an element of $\mathbb{H}$, denoted by $\int \psi d\nu$, such that $\langle h, \int \psi d\nu \rangle = \int \langle h, \psi(y) \rangle \nu(dy)$ for all $h \in \mathbb{H}$.*

Proof. For all $h \in \mathbb{H}$ and $y \in \mathbb{Y}$, $y \mapsto \langle h, \psi(y) \rangle$ is measurable by Lemma 1, and $|\langle h, \psi(y) \rangle| \leq \|h\| \|\psi(y)\| = \|h\| \sqrt{k(y, y)}$. Therefore, for all $\nu \in \mathbb{M}_H$ and all $h \in \mathbb{H}$, it holds that

$$\begin{aligned} \int |\langle h, \psi(y) \rangle| d|\nu| &\leq \|h\| \int \sqrt{k(y, y)} |\nu|(dy) \\ &\leq \|h\| (|\nu|(\mathbb{Y}))^{\frac{1}{2}} \left(\int k(y, y) |\nu|(dy) \right)^{\frac{1}{2}} < +\infty. \end{aligned}$$

As a consequence, the linear functional $\mathbb{H} \rightarrow \mathbb{R}$, $h \mapsto \int \langle h, \psi(y) \rangle \nu(dy)$ is well-defined and continuous, and the existence of the Pettis integral $\int \psi d\nu$ follows from the Riesz representation theorem. $\square$

Remark 16. If $\mathbb{H}$ is an RKHS and $\psi(y) = k(y, \cdot)$ as in Remark 15, then for all $\nu \in \mathbb{M}_H$, we have $\int k(y, y) \nu(dy) = \int \|\psi\|^2 d\nu < +\infty$, and the Pettis-integral $\int \psi d\nu$ coincides with the kernel mean embedding $\int k(\cdot, y) \nu(dy)$, where the integral is now defined in the Lebesgue sense.

Proof of Proposition 3. Let $\rho : \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$ be symmetric, measurable and vanishing on the diagonal. If ρ is CIND, then it is also CND, since finitely supported measures belong to $\mathbb{M}_H$. Conversely, assume that ρ is CND. Let $\mathbb{H}$ and ψ be as in Theorem 7. Then $\int \psi d\nu$ exists in the Pettis sense by Lemma 5, for all $\nu \in \mathbb{M}_H$. Assume in addition that $\nu(\mathbb{Y}) = 0$. Then we have

$$\iint \rho d(\nu \otimes \nu) = \iint \|\psi(y) - \psi(y')\|^2 \nu(dy) \nu(dy')$$

$$\begin{aligned}
&= \iint \left[\|\psi(y)\|^2 + \|\psi(y')\|^2 - 2 \langle \psi(y), \psi(y') \rangle \right] \nu(dy) \nu(dy') \\
&= \int \left[0 + \int \|\psi(y')\|^2 \nu(dy') - 2 \langle \psi(y), \int \psi d\nu \rangle \right] \nu(dy) \\
&= -2 \left\| \int \psi d\nu \right\|^2 \leq 0,
\end{aligned}$$

which proves that ρ is CIND. $\square$

A.3 Proof of Theorem 6

For all $m \in \mathbb{N}^*$, we denote by Σ_{m-1} the $(m-1)$ -dimensional probability simplex in $\mathbb{R}^m$, that is, $\Sigma_{m-1} = \{\nu \in \mathbb{R}^m \mid \sum_{i=1}^m \nu_i = 1, \nu_i \geq 0, \text{ for } i = 1, \dots, m\}$. Note that the letter ν , which denotes elsewhere in the paper a probability measure, is used in this section to denote an element of the probability simplex. The relative interior of Σ_{m-1} , denoted by $\text{ri}(\Sigma_{m-1})$, consists of all probability vectors with strictly positive components, namely, $\text{ri}(\Sigma_{m-1}) = \{\nu \in \mathbb{R}^m \mid \sum_{i=1}^m \nu_i = 1, \nu_i > 0, \text{ for } i = 1, \dots, m\}$.

Lemma 6. *Let $m \in \mathbb{N}^*$. Let $H : \Sigma_{m-1} \rightarrow \overline{\mathbb{R}}_+$ denote a concave function on the probability simplex, taking non-negative extended-real values. Then there exists a set $\Phi \subset \overline{\mathbb{R}}_+^m$ such that i) for all $\nu \in \Sigma_{m-1}$,*

$$H(\nu) \leq \inf_{\varphi \in \Phi} \langle \varphi, \nu \rangle, \quad (\text{A4})$$

and ii) for all $\nu \in \text{ri}(\Sigma_{m-1})$, the infimum is attained and the inequality is an equality.

Proof. Assume first that $H(\nu) = +\infty$ for at least one $\nu \in \Sigma_{m-1}$. Then, by concavity, $H(\nu) = +\infty$ for all $\nu \in \text{ri}(\Sigma_{m-1})$. As a consequence, the claim holds with $\Phi = \{+\infty \mathbb{1}\}$, with $\mathbb{1}$ denoting the constant function over the simplex taking the value 1.

Assume now that $H(\nu) < +\infty$ for all $\nu \in \Sigma_{m-1}$. It is always true that

$$H(\nu) \leq \inf_a a(\nu), \quad (\text{A5})$$

where the infimum runs over all affine majorants of H . Moreover, it is a well-known fact from convex analysis that, since H is concave, then on the relative interior $\text{ri}(\Sigma_{m-1})$ the function is continuous, the inequality (A5) is an equality, and the infimum in the right-hand side is attained. To conclude the proof, observe that for $a : \nu \mapsto \langle \alpha, \nu \rangle + \beta$ we have $a(\nu) = \langle \alpha + \beta \mathbb{1}, \nu \rangle$ for all $\nu \in \Sigma_{m-1}$, and the components of $\alpha + \beta \mathbb{1}$ are non-negative since $\alpha_j + \beta = \langle \alpha + \beta \mathbb{1}, e_j \rangle \geq H(e_j) \geq 0$ for all $j \in \{1, \dots, m\}$, where e_j is the j -th element of the canonical basis of $\mathbb{R}^m$ (corresponding to a Dirac measure on the j -th element of $\mathbb{Y}$). $\square$

Proof of Theorem 6. Let $m \in \mathbb{N}^*$ denote the cardinal of $\mathbb{Y}$, and let $y_1, \dots, y_m$ denote the elements of $\mathbb{Y}$. Assume without loss of generality that $\mathcal{Y} = \mathcal{P}(\mathbb{Y})$ —otherwise, we simply need to work with the atoms of $\mathcal{Y}$ instead of $\mathbb{Y}$ itself. Identify $\mathbb{P}$ with the $(m-1)$ -dimensional probability simplex in $\Sigma_{m-1} \subset \mathbb{R}^m$, and H with a non-negative, concave function on Σ_{m-1} . The integral $\int L(y, d) \nu(dy)$ in the right-hand side of (16) becomes, through this identification, the (extended) scalar product $\langle \varphi_d, \nu \rangle = \sum_{k=1}^m \varphi_{d,k} \nu_k$, where $\varphi_d = (L(y_k, d))_{1 \leq k \leq m} \in \mathbb{R}^m$ and the convention $(+\infty) \cdot 0 = 0$ is used when $\varphi_{d,k} = +\infty$ and $\nu_k = 0$. The claim to be proved can thus be reformulated as

$(P_m) :$ for all non-negative, concave functions $H : \Sigma_{m-1} \rightarrow \overline{\mathbb{R}}_+$, there exists a set

$\Phi \subset \overline{\mathbb{R}}_+^m$ such that, for all $\nu \in \Sigma_{m-1}$, $H(\nu) = \min_{\varphi \in \Phi} \langle \varphi, \nu \rangle$.

We prove the result by induction on $m \in \mathbb{N}^*$. For $m = 1$, $\Sigma_{m-1} = \Sigma_0 = \{1\}$, and therefore P_1 trivially holds with $\Phi = \{H(1)\}$. Assume now that P_m holds for some $m \in \mathbb{N}^*$. Let $H : \Sigma_m \rightarrow \overline{\mathbb{R}}_+$ be concave. By Lemma 6, there exists $\Phi_0 \subset \overline{\mathbb{R}}_+^{m+1}$ such that

$$H(\nu) \leq \inf_{\varphi \in \Phi_0} \langle \varphi, \nu \rangle, \quad (\text{A6})$$

for all $\nu \in \Sigma_m$, with equality and the infimum attained on $\text{ri}(\Sigma_m)$. Let $F_1, \dots, F_{m+1}$ denote the $(m-1)$ -dimensional faces of Σ_m , where $F_k = \{\nu \in \Sigma_m \mid \nu_k = 0\}$. Then $\Sigma_m \setminus \text{ri}(\Sigma_m) = \bigcup_{k=1}^{m+1} F_k$, and each F_k is linearly isomorphic to Σ_{m-1} . More precisely, we have $F_k = A_k \Sigma_{m-1}$, where $A_k \in \mathbb{R}^{(m+1) \times m}$ is the matrix with entries in $\{0, 1\}$ such that $\nu \mapsto A_k \nu$ inserts a zero in the k -th position, and $\nu \mapsto A_k^\top \nu$ is the inverse mapping. Thus, for each $k \in \{1, \dots, m+1\}$, $H_k^0 : \nu \mapsto H(A_k \nu)$ is a concave function on Σ_{m-1} , and therefore by P_m there exists a set $\Phi_k^0 \subset \overline{\mathbb{R}}_+^m$ such that $H_k^0(\nu) = \min_{\varphi \in \Phi_k^0} \langle \varphi, \nu \rangle$, for all $\nu \in \Sigma_{m-1}$. As a consequence, for all $\nu \in F_k$ we have

$$H(\nu) = H_k^0(A_k^\top \nu) = \min_{\varphi \in \Phi_k^0} \langle \varphi, A_k^\top \nu \rangle = \min_{\varphi \in \Phi_k^0} \langle A_k \varphi, \nu \rangle. \quad (\text{A7})$$

Set $\Phi_k = \{\varphi \in \overline{\mathbb{R}}_+^{m+1} \mid \exists \varphi^0 \in \Phi_k^0, \varphi = A_k \varphi^0 + (+\infty) e_k\}$, where e_k is the k -th element of the canonical basis. Then, for all $\nu \in \Sigma_m$, we have

$$\min_{\varphi \in \Phi_k} \langle \varphi, \nu \rangle = \begin{cases} H(\nu) & \text{if } \nu \in F_k, \\ +\infty & \text{otherwise.} \end{cases}$$

Setting $\Phi = \Phi_0 \cup \Phi_1 \cup \dots \cup \Phi_{m+1}$ proves P_{m+1} . The infimum is attained for some $\varphi \in \Phi_0$ for $\nu \in \text{ri}(\Sigma_m)$, and for some $\varphi \in \Phi_k$ is $\nu \in F_k \subset \Sigma_m \setminus (\text{ri}(\Sigma_m))$. $\square$

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